

Software Architecture

Introduction

BSc



Ingo Arnold

Module Opening

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Education		Publications	
Computer Science (Informatik)		Books • Enterprise Architecture Function — a pattern language for planning, building, and executing (2022) • Software Architecture — a comprehensive framework and guide for practitioners (2011) • Software-Architektur — Grundlagen Konzepte Praxis (2005, 2008)	
Work History		Papers & Forum Speeches • Ingo Arnold in	
Enterprise and Solution Architecture Consulting GmbH – CEO (5 years) Novartis — Head Architecture Management (18 years) IBM — IT Architect (3 years) Ciba Geigy — Head Business Solutions (5 years)			
Lecturing			
@ University of Basel • Software Architecture • Software Life-Cycle Management		@ University of Cooperative Education, Lörrach • Object-Oriented Programming • Advanced Software-Engineering • Software Design Patterns	
@ University of Applied Sciences North-Western Switzerland • Software Architecture (BSc) • Managing IT in a digitized World – IT Architecture Module (MSc) • IT Management (CAS)			
Miscellaneous			
Family • 2 sons (both studied computer science)		Sports • Climbing, Mountainbiking, Snowboarding	Other • Curious, broadly interested, nature lover, appreciate good books, deep and open conversations, ...

Module Opening

Motivation

Your ...

- Name
- Experience and background
- Software architecture conception and expectations
- ...

Go to **<https://menti.com>** and enter the code on the screen.

Module Opening

Few Things up-front – Logistics

- **Room and time:** Room as communicated. Time: 08:15 to 11:00 / 12:15 to 15:00.
- We will usually have 15 minute breaks every 45 minutes (i.e., two per lecture).
- Your **active participation** and questions are prerequisites for your learning success.
- Basic knowledge of the **object-oriented paradigm**, the **Java** programming language, and software **design patterns** are prerequisites for this course.
- I do **not prescribe tools** that we use. However, examples are mostly based on the Eclipse IDE.
- All lecture handouts (mostly PDF files) will be regularly uploaded to the **lecture decks location**.

Module Opening

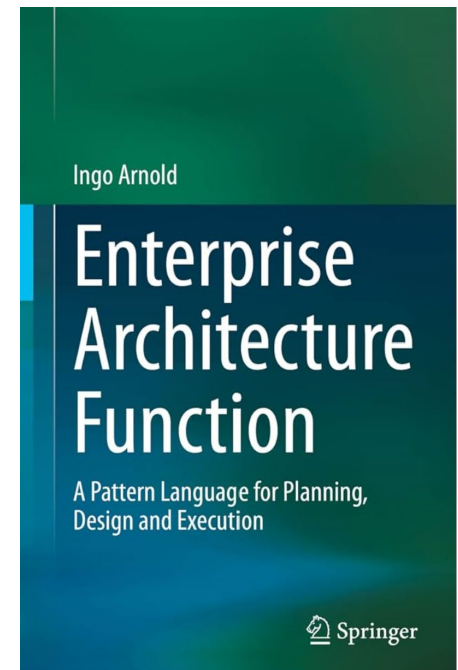
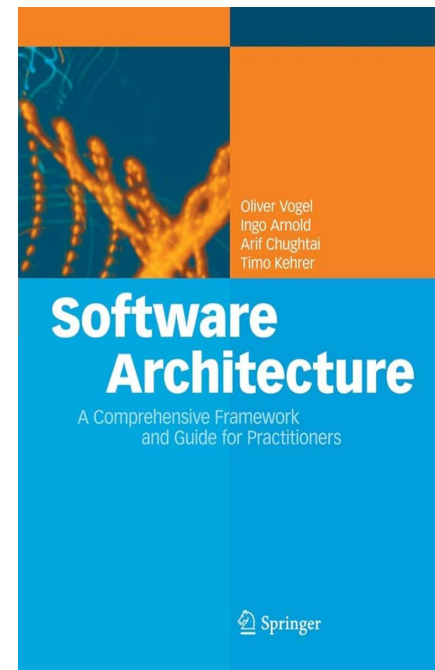
Few Things up-front – Learning (Approach)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!**
 - The **initial lecture** is an **exception** because I **will guide** you **through my** set of **slides**.
 - For **all subsequent lectures**, **you will work through** a set of **slides** containing relevant software architecture knowledge **at home** so that **you come prepared** to the **next lecture**.
 - **During class**, **we will jointly apply** what **you have learned at home**.
 - The **theoretical knowledge slide sets** form the **core** of my **teaching materials**.
 - What I have not done (**not managed to do**) to date is to **create videos** in which I explain the slide set material in more detail.
 - However, I have **structured the slide sets** so that they can also **serve as a script** in addition to providing clear illustrations. In this respect, the **slide sets** are sufficiently **self-explanatory**.

Module Opening

Few Things up-front – Learning (Approach)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - In **addition to the slide sets**, you will **receive further instructions** for **working through** individual **slide set chapters** in the form of **accompanying documents** ('_Instruction').
 -  1.1-BSc_SWA_WHAT_Concepts.pptx
 -  1.1-BSc_SWA_WHAT_Concepts_Instruction.docx
 -  1.1-BSc_SWA_WHAT_Concepts_Instruction.pdf
 - The **accompanying document** ('_Instruction') also **contains questions** to answer and **literature references** that refer you to further sources of information.
 - **Crucial resources** I'll be referring heavily are the **two books to the right**.



Module Opening

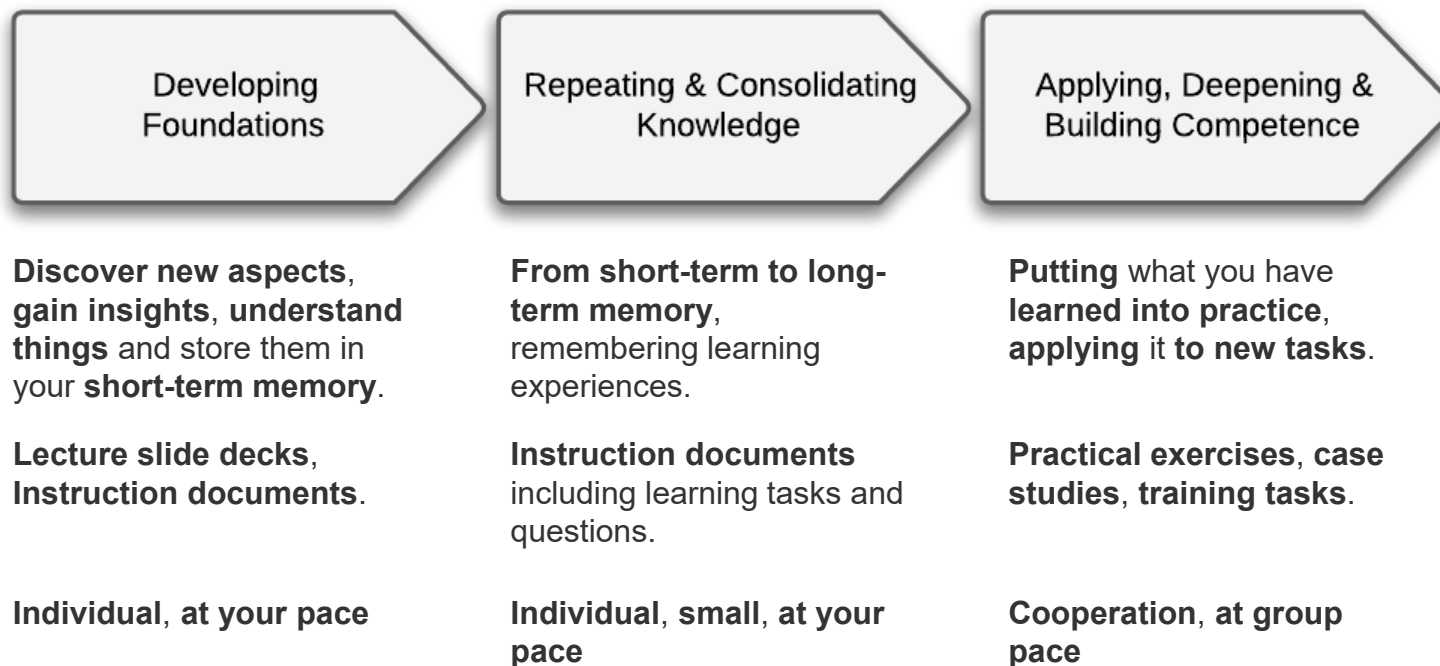
Few Things up-front – Learning (Approach)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - **You are** of course also **welcome to use ChatGPT** or similar tools at home to help you learn the theoretical material covered in the slide set in a more personalised way.
 - Further, **you will receive literature lists** at the **end of each slide deck**.
 - **If** you still **have questions** after working through an appropriate slide set, please make a note of them and **bring them with you to the next lecture**.
 - **I will answer** any **open questions** at the **beginning of each lecture** and **before moving on** to the **application of what you have just learned**.

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Few Things up-front – Learning (Approach)

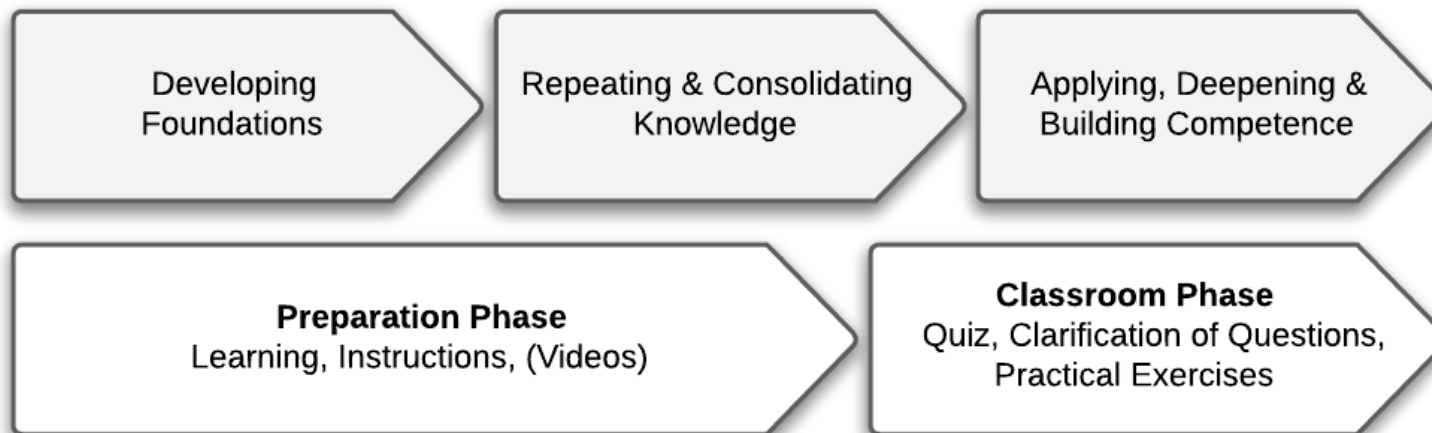
- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - The learning concept behind the revised lecture is described below.



Module Opening

Few Things up-front – Learning (Approach)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - The learning concept behind the revised lecture is described below.



Module Opening

Few Things up-front – Learning (Approach)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - Preparation phase misconceptions
 - Preparation $\neq$ Scrolling through PPT decks.
 - Tasks and questions in the Instruction documents are essential elements.
 - Ask questions – even during the preparation phase.
 - The aim of preparation is not a perfect understanding, but rather being able to participate in the classroom phase.
 - If you are unable to participate in a unit (e.g. due to illness), join in with the current topic and catch up on what you have missed later.

Module Opening

Few Things up-front – Learning (Time Management)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - Pay attention to what you do and how long you do something!
 - If you get stuck, break out of your inner loop and start asking questions
 - Use discussion forums in Teams, chat, email, etc.
 - Form groups and support each other
 - Collect questions for the classroom phase
 - Don't be a perfectionist when it comes to learning tasks
 - It's (often) not about finding the (perfect) solution
 - It's (also) about thinking things through to understand a given problem and see a solution shine through it
 - It's about learning from experience, including what doesn't work ('those who fail productively learn more')

Module Opening

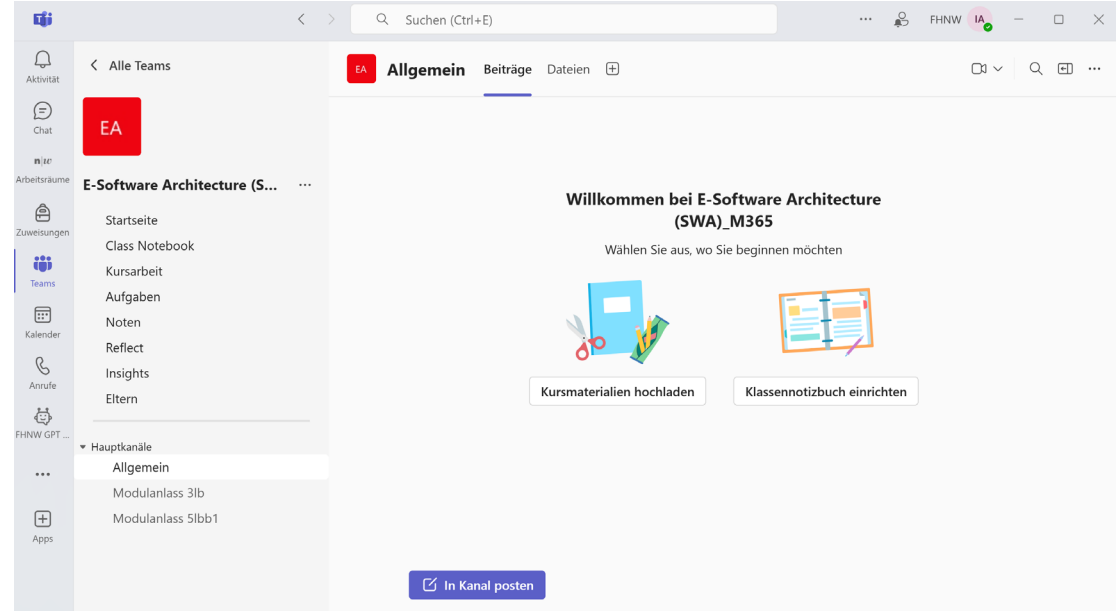
Few Things up-front – Learning (Failure)

- I have **completely revamped** the **Software Architecture lecture** for this semester. In this respect, **this semester is a premium!** (continued)
 - “The greatest teacher, failure is.” (Yoda)
 - “Negative results are just what I want. They’re just as valuable to me as positive results. I can never find the thing that does the job best until I find the ones that don’t.” (Thomas Edison)
 - “Wir werden Fehler feiern!” (Christian Spannagel)

Module Opening

Few Things up-front – Learning (MS Teams)

- One Channel per module
 - Use them to conduct discussions and ask questions.
 - Help others by answering questions.
- Chat/call for study groups in the preparation phase
- Channel “General” – under “Files”
 - Slide decks
 - Instruction documents
 - Exercises



Module Opening

Few Things up-front – Exam

- Exam
 - At the end of the semester, you will write an **exam**.
 - At the same time, you will write a **preliminary exam** in the middle of the semester, 50% of which will be included in the final semester grade.



Module Opening

Learning Objectives

You ...

- know the **architecture elaboration process** for systematically developing a solution to a given problem
- can **analyze requirements** with respect to their architectural significance and ensure traceability between requirement and design
- know important **architectural views** on the design of software systems
- are able to **represent the design of systems** in architectural views and recognize architectural decisions coded in them
- can **justify own design** decisions and develop own system architectures
- can **evaluate system architectures** as a whole and in parts with respect to their advantages and disadvantages

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Cornerstone Systems and Information

My contact details

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- eMail: ingo.arnold@fhnw.ch (ingo.e.arnold@outlook.com)
- Webex <https://fhnw.webex.com/meet/ingo.arnold>

Lecture decks location

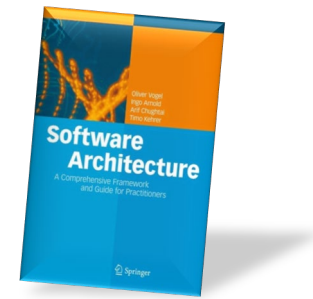
FHNW-share

IT Support

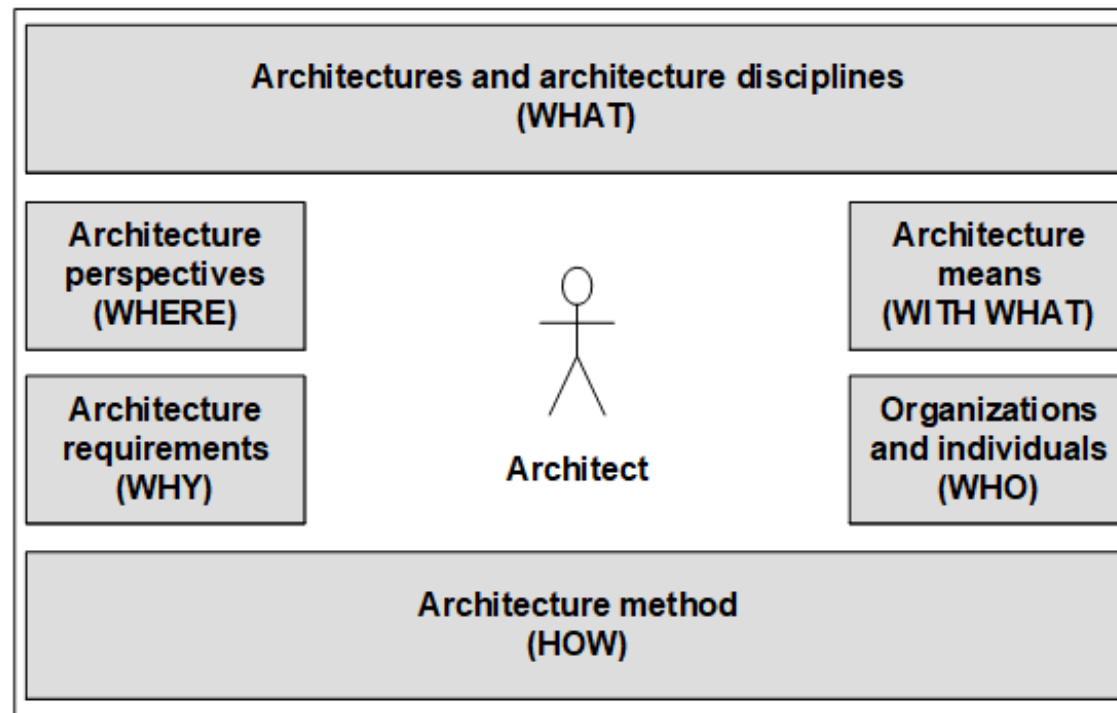
it-support@fhnw.ch / +41 61 228 50 50

Module Opening

Structure and Order

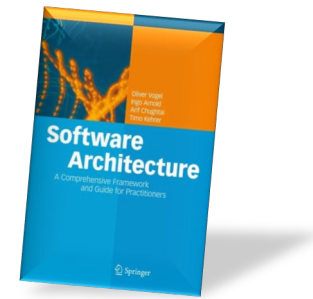


The high-level structure of the lecture is based on the book [Software Architecture: A Comprehensive Framework and Guide for Practitioners](#) [Vogel, Arnold et al 2011].

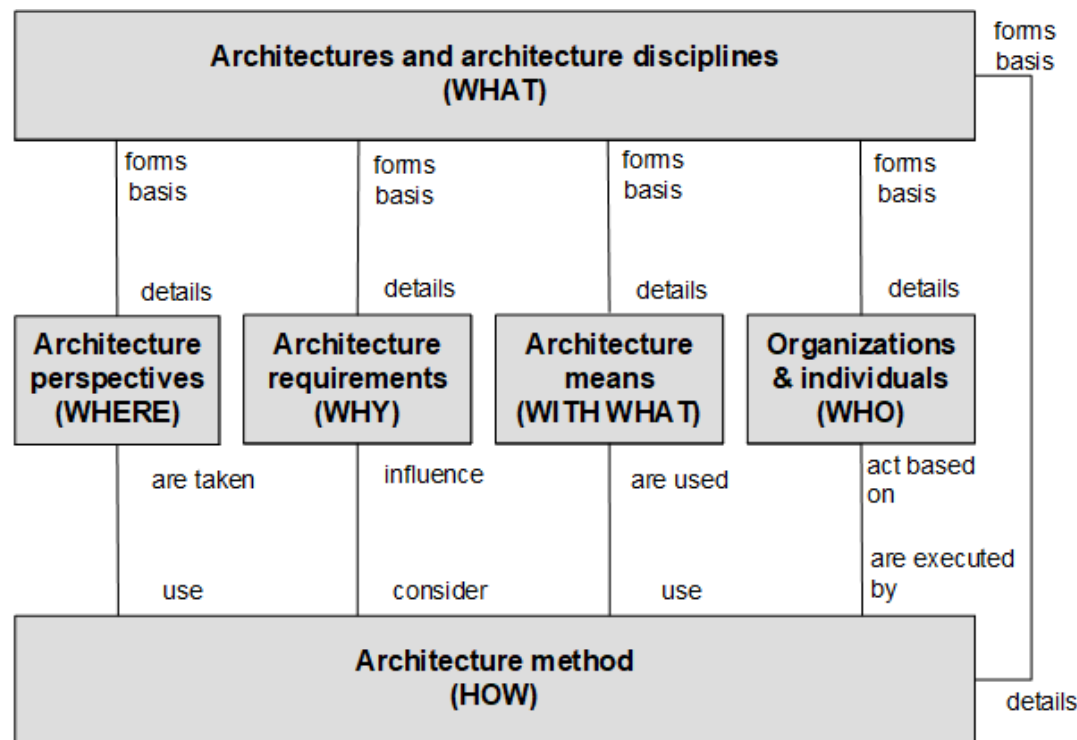


Module Opening

Structure and Order



The strict differentiation of dimensions and their isolated consideration is only useful for rough orientation purposes. **Software architecture is a holistic discipline** and this entirety is lost when it is considered too particularized.

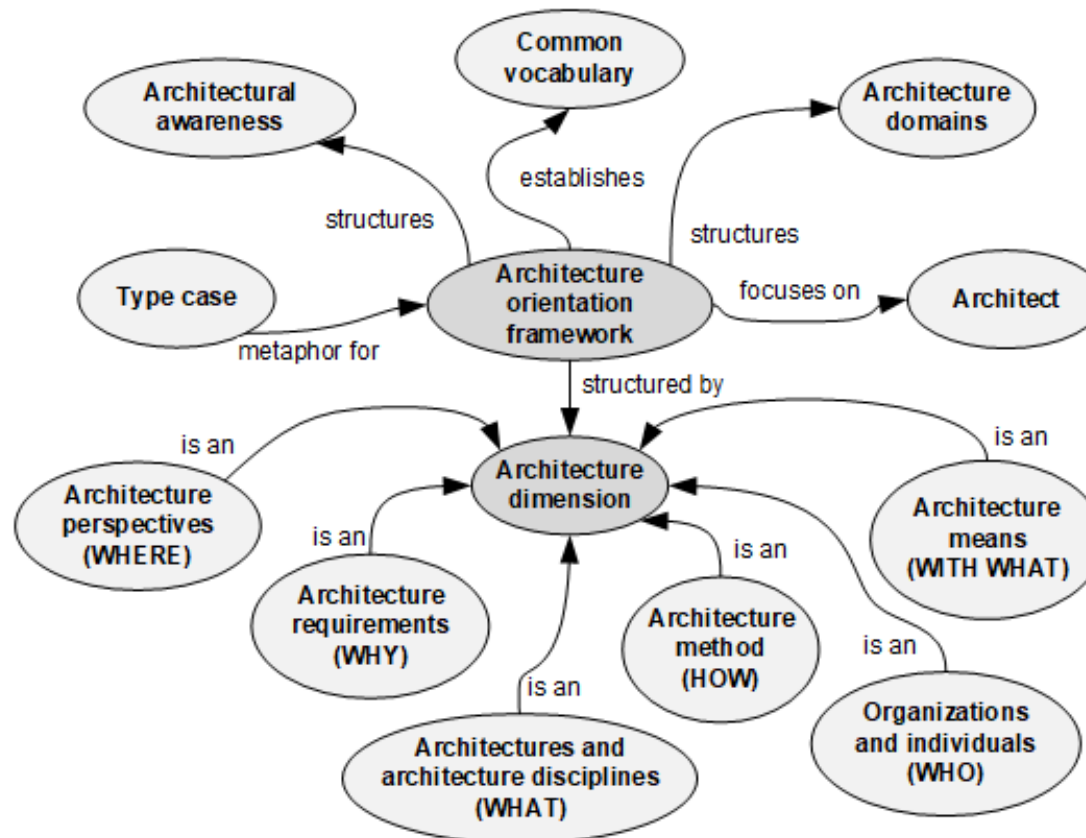


Module Opening

Structure and Order



Nevertheless, we will use the dimensions as an overarching orientation framework and place short introductions to each dimension at its beginning.



Questions



Bibliography

Lecture

[Arnold 2022]

Arnold, Ingo, *Enterprise Architecture Function — a pattern language for planning, designing, and executing*, Springer Science and Business Media, Berlin Heidelberg, 2022

[Vogel, Arnold et al 2011]

Vogel, Oliver; Arnold, Ingo; Chugtai, Arif; Kehrer, Timo, *Software Architecture: A Comprehensive Framework and Guide for Practitioners*, Springer Science and Business Media, Berlin Heidelberg, 2011